

Lecture One

1 Matrices and its operations

1.1 Matrices

Definition 1.1.1 A matrix A is a rectangular array of scalars usually presented in the following form:

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The row of such a matrix A are the m horizontal lists of scalars:

$$(a_{11}, a_{12}, \dots, a_{1n}), (a_{21}, a_{22}, \dots, a_{2n}), \dots, (a_{m1}, a_{m2}, \dots, a_{mn})$$

and the columns of A are the n vertical lists of scalars:

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Note that:

1. The element a_{ij} , called the ij -entry or ij -element appears in row i and column j .
2. A matrix with m rows and n columns is called an m by n matrix and written as $(m \times n)$.
3. The pair of numbers m and n is called the **size of the matrix**.
4. A matrix with only one row is called a **row matrix** or **row vector**.
5. A matrix with only one column is called a **column matrix** or **column vector**.
6. A matrix whose entries are all zero is called a **zero matrix** and will usually denoted by 0.
7. Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ **are equal** if and only if they have the same size $(m \times n)$ and $a_{ij} = b_{ij}$ for $1 \leq i \leq m$ and $1 \leq j \leq n$.

Example 1.1.1 (a) The rectangular array $A = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & 2 \end{bmatrix}$ is 2×3 matrix. Its rows are $(1, -4, 5)$ and $(0, 3, -2)$ and its columns are $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ -2 \end{bmatrix}$

(b) The 2×4 zero matrix is the matrix $0 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

(c) Find x, y, z, w such that:

$$\begin{bmatrix} x+y & 2z+w \\ x-y & z-w \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 5 \end{bmatrix}$$

By definition of equality of matrices, the four corresponding entries must be equal. Thus:

$$\left. \begin{matrix} x+y=3 \\ x-y=1 \end{matrix} \right\} \Rightarrow x=2, y=1 \quad \& \quad \left. \begin{matrix} 2z+w=3 \\ z-w=1 \end{matrix} \right\} \Rightarrow z=4, w=-1$$

(d) Find x, y, z, t where:

$$\begin{bmatrix} 3x & 3y \\ 3z & 3t \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2t \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+t & 3 \end{bmatrix}$$

1.2 Matrices Addition and Scalar Multiplication

Definition 1.2.2 Definition of Matrices Addition: Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices with the same size, say $m \times n$ matrices. The sum of A and B written $A + B$, is the matrix obtained by adding corresponding elements from A and B . That is:

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{bmatrix}$$

Definition 1.2.3 The product of the matrix A by a scalar k , written $k \times A$ or simply kA , is the matrix obtained by multiplying each element of A by k . That is:

$$kA = \begin{bmatrix} k \cdot a_{11} & k \cdot a_{12} & \cdots & k \cdot a_{1n} \\ k \cdot a_{21} & k \cdot a_{22} & \cdots & k \cdot a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k \cdot a_{m1} & k \cdot a_{m2} & \cdots & k \cdot a_{mn} \end{bmatrix}$$

Note that:

1. $A + B$ and kA are also $m \times n$ matrices. We also define $-A = (-1)A$ and $A - B = A + (-B)$.
2. The matrix $-A$ is called the negative of the matrix A (Inverse addition), and the matrix $A - B$ is called the difference of A and B .
3. The sum of matrices with different sizes is not defined.

Example 1.2.2 Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 6 & 8 \\ 1 & -3 & -7 \end{bmatrix}$, then:

$$(i) \ A + B = \begin{bmatrix} 1+4 & (-2)+6 & 3+8 \\ 0+1 & 4+(-3) & 5+(-7) \end{bmatrix} = \begin{bmatrix} 5 & 4 & 11 \\ 1 & 1 & -2 \end{bmatrix}$$

$$(ii) \ 3A = 3 \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 3(1) & 3(-2) & 3(3) \\ 3(0) & 3(4) & 3(5) \end{bmatrix} = \begin{bmatrix} 3 & -6 & 9 \\ 0 & 12 & 15 \end{bmatrix}$$

$$(iii) \ 2A - 3B = 2A + (-3B) = \begin{bmatrix} 2 & -4 & 6 \\ 0 & 8 & 10 \end{bmatrix} + \begin{bmatrix} -12 & -18 & -24 \\ -3 & 9 & 21 \end{bmatrix} = \begin{bmatrix} -10 & -22 & -18 \\ -3 & 17 & 31 \end{bmatrix}$$

The matrix $2A - 3B$ is called a linear combination of A and B .

1.3 Properties of matrices under the operation of matrices addition and scalar multiplication

Theorem 1.3.1 Consider A, B and C are 3 matrices (with the same size) and any scalars L, M . Then:

1. $(A + B) + C = A + (B + C)$
2. $A + 0 = 0 + A = A$
3. $A + (-A) = (-A) + A = 0$
4. $A + B = B + A$
5. $L(A + B) = LA + LB$
6. $(L + M)A = LA + MA$
7. $(LM)A = L(MA)$
8. $1.A = A$

1.4 Matrices Multiplication

Definition 1.4.4 If $A = [a_{ij}]$ is an $m \times n$ matrix, and $B = [b_{ij}]$ is an $n \times p$ matrix, then the product AB is an $m \times p$ matrix in the form

$$AB = [c_{ij}] \text{ where } c_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{in}b_{nj}$$

This definition means that the entry in the i th row and the j th column of the product AB is obtained by multiplying the entries in the i th row of A by the corresponding entries in the j th column of B and then adding the result. The next example illustrates this process.

Example 1.4.3 Find the product AB , where $A = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix}$.

Sol

Firstly: note that the product AB is defined because A has size 3×2 and B has size 2×2 . Moreover, the product AB has 3×2 and will take the form:

$$\begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix}$$

Such that:

$$c_{11} = [-1 \ 3] \begin{bmatrix} -3 \\ -4 \end{bmatrix} = (-1)(-3) + (3)(-4) = -9$$

$$c_{12} = [-1 \ 3] \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (-1)(2) + (3)(1) = 1$$

$$c_{21} = [4 \ -2] \begin{bmatrix} -3 \\ -4 \end{bmatrix} = (4)(-3) + (-2)(-4) = -4$$

$$c_{22} = [4 \ -2] \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (4)(2) + (-2)(1) = 6$$

$$c_{31} = [5 \ 0] \begin{bmatrix} -3 \\ -4 \end{bmatrix} = (5)(-3) + (0)(-4) = -15$$

$$c_{32} = [5 \ 0] \begin{bmatrix} 2 \\ 1 \end{bmatrix} = (5)(2) + (0)(1) = 10$$

$$\text{Thus the product } AB = \begin{bmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{bmatrix}$$

Example 1.4.4 1. $\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 2 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} -5 & 7 & -1 \\ -3 & 6 & 6 \end{bmatrix}$.

2. $\begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ -2 & 5 \end{bmatrix}$.

3. $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

4. $[1 \ -2 \ -3] \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = [1]$.

5. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 0 & -2 \end{bmatrix}$, then $AB = \begin{bmatrix} 5 & 2 \\ 15 & 10 \end{bmatrix}$ and $BA = \begin{bmatrix} 23 & 34 \\ -6 & -8 \end{bmatrix}$. This example shows that matrices multiplication is not commutative, that is in general $AB \neq BA$. However matrices multiplication does satisfy the following properties.

Theorem 1.4.2 Let A, B, C be matrices. Then whenever the products and sums are defined:

1. $(AB)C = A(BC)$ (Associative law).
2. $A(B + C) = AB + AC$ (Left distributive law).
3. $(B + C)A = BA + CA$ (Right distributive law).
4. $k(AB) = (kA)B = A(kB)$ where k is a scalar.

Note that $0A = 0$ and $B0 = 0$ where 0 is the zero matrix.

1.5 Transpose of Matrix

Definition 1.5.5 The transpose of a matrix A , Written A^T is the matrix obtained by writing the columns of A , in order as rows.

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & -3 & -5 \end{bmatrix}^T = \begin{bmatrix} 1 \\ -3 \\ -5 \end{bmatrix}$$

In other words, if $A = [a_{ij}]$ is an $m \times n$ matrix, then $A^T = [b_{ij}]$ is the $n \times m$ matrix where $b_{ij} = a_{ji}$.

Observe that the transpose of a row vector is a column vector. Similarly, the transpose of a column vector is a row vector.

Theorem 1.5.3 Let A and B be matrices and let k be a scalar. Then, whenever the sum and product are defined:

- (i) $(A + B)^T = A^T + B^T$
- (ii) $(A^T)^T = A$
- (iii) $(kA)^T = kA^T$
- (iv) $(AB)^T = B^T A^T$

1.6 Diagonal and Trace

Definition 1.6.6 Let $A = [a_{ij}]$ be an n -square matrix. The **diagonal** or **main diagonal** of A consists of the elements with the same subscripts - that is $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$.

Definition 1.6.7 The **trace** of A , written $\text{tr}(A)$, is the sum of the diagonal elements. Namely

$$\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

Theorem 1.6.4 Suppose $A = [a_{ij}]$ and $B = [b_{ij}]$ are n -square matrices and k is a scalar. Then:

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
2. $\text{tr}(A^T) = \text{tr}(A)$
3. $\text{tr}(kA) = k \text{tr}(A)$
4. $\text{tr}(AB) = \text{tr}(BA)$

1.7 Identity Matrix

Definition 1.7.8 The n -square identity or unit matrix, denoted by I_n or simply I , is the n -square matrix with 1's on the diagonal and 0's elsewhere.

The identity matrix I is similar to the scalar 1, in that for any n -square matrix A

$$AI = IA = A$$

and we can write it as:

$$I_n = [a_{ij}] = \begin{cases} 0 & , \text{if } i \neq j \\ 1 & , \text{if } i = j \end{cases}$$

1.8 Triangular Matrices

1. **Definition 1.8.9 Upper Triangular Matrix:** A square matrix $A = [a_{ij}]$ is upper triangular if all entries below the main diagonal are equal to 0.

For example:

$$\begin{bmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{bmatrix}, \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ 0 & b_{22} & b_{23} \\ 0 & 0 & b_{33} \end{bmatrix}, \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ 0 & c_{22} & c_{23} & c_{24} \\ 0 & 0 & c_{33} & c_{34} \\ 0 & 0 & 0 & c_{44} \end{bmatrix}$$

2. **Definition 1.8.10 Lower Triangular Matrix:** A square matrix $A = [a_{ij}]$ is lower triangular if all entries above the main diagonal are equal to 0.

For example:

$$\begin{bmatrix} a_{11} & 0 \\ a_{21} & a_{22} \end{bmatrix}, \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix}, \begin{bmatrix} c_{11} & 0 & 0 & 0 \\ c_{21} & c_{22} & 0 & 0 \\ c_{31} & c_{32} & c_{33} & 0 \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix}$$

1.9 Symmetric Matrices

Definition 1.9.11 A matrix A is **symmetric** if $A^T = A$. Equivalently, $A = [a_{ij}]$ is symmetric elements (mirror elements) with respect to the diagonal are equal-that is, if each $a_{ij} = a_{ji}$.

Definition 1.9.12 A matrix A is **Skew symmetric** if $A^T = -A$, or equivalently if each $a_{ij} = -a_{ji}$

Note: Clearly, the diagonal elements of such a matrix must be zero, because $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$

1.10 Powers of Matrices, Polynomials in Matrices

Let A be an n -square matrix, powers of A are defined as follows:

$$A^2 = A.A, A^3 = A^2.A, \dots, A^{n+1} = A^n.A \quad \text{and} \quad A^0 = I$$

Polynomials in the matrix A are also defined, specifically for any polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

Where the a_i are scalars, $f(A)$ is defined to be the following matrix.

$$f(A) = a_0I + a_1A + a_2A^2 + \dots + a_nA^n$$

Note:

1. $f(A)$ is obtained from $f(x)$ by substituting the matrix A for the variable x and substituting the scalar matrix a_0I for the scalar a_0 .
2. If $f(A)$ is the zero matrix, then A is called a zero or root of $f(x)$.

Example 1.10.5 Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix}$ find A^2, A^3 , then suppose $f(x) = 2x^2 - 3x + 5$ and $g(x) = x^2 + 3x - 10$, find $f(A), g(A)$.

Sol

$$A^2 = A.A = \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix}$$

$$A^3 = A^2.A = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} = \begin{bmatrix} -11 & 38 \\ 57 & -106 \end{bmatrix}$$

Now:

$$f(A) = 2 \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} - 3 \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} + 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 16 & -18 \\ -27 & 61 \end{bmatrix}$$

$$g(A) = \begin{bmatrix} 7 & -6 \\ -9 & 22 \end{bmatrix} + 3 \begin{bmatrix} 1 & 2 \\ 3 & -4 \end{bmatrix} - 10 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus A is a zero of the polynomial $g(x)$.